BubbleVision: An Automated Bubble Test Checker and Item Analysis Web App using K-Nearest Neighbors

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I. INTRODUCTION

Assessment is crucial in education because it offers teachers important information about student learning and the overall success of teaching methods. By utilizing different assessment techniques, educators can ascertain if students have reached the desired competencies. A key element of this process is item analysis, which assesses the quality of test questions by pinpointing items that might not accurately gauge student learning. Results from item analysis assist educators in honing their test questions, enhancing teaching methods, and confirming that evaluations correctly represent student comprehension.

A standard paper-based exam comprises two main parts: the test questionnaire and the bubble-style answer sheet. The questionnaire includes the real test questions, whereas the answer sheet features an organized layout of circles (bubbles) that students fill in to indicate their choices. This layout enables quicker verification and more efficient gathering of responses. Nonetheless, correctly interpreting bubble sheets typically necessitates Optical Mark Reader (OMR) devices, which depend on exact alignment and specific paper types. These gadgets are costly, need upkeep, and are frequently inaccessible in schools with limited resources.

As a result, numerous educators continue to depend on manual verification and manual item analysis. They count student answers one by one, match them with the answer key, calculate scores, and determine item easy values. This manual procedure takes a lot of time, is susceptible to mistakes, and is difficult for large groups of students. Consequently, there is an increasing demand for an affordable, dependable, and effective system capable of automating bubble sheet processing and item analysis without the need for specialized equipment.

Progress in machine learning (ML) has made automated systems more feasible. The K-Nearest Neighbors (KNN) algorithm is a straightforward and efficient classification method, particularly useful for determining if a bubble is shaded, unshaded, or invalid. Its low computational requirements make it an economical choice for schools with limited resources. Previous research has demonstrated KNN's effectiveness in visual classification tasks, such as digit recognition in OCR systems, making it ideal for automating bubble sheet grading (Lu, 2024).

This research introduces BubbleVision: An Automated Bubble Test Checker and Item Analysis System that uses the K-Nearest Neighbors (KNN) algorithm. The system analyzes captured or uploaded bubble sheets, classifies student responses using KNN, compares them with the answer key, and generates a detailed item analysis report. Based on the item index, the system automatically marks and categorizes items as easy, right difficult, or difficult.

This research ultimately seeks to enhance the accessibility, accuracy, and efficiency of bubble sheet checking and item analysis, especially for schools lacking the resources for conventional OMR technology. Using readily accessible tools like smartphone cameras and or uploaded photo, the system promotes evidence-based instruction, ongoing enhancement of assessment items, and improved learning results for students

Statement of the Problem

Item analysis is essential in education for reliable assessments. It helps teachers identify question quality, yet many schools still rely on manual processes to check responses and compute results. This method is slow and error-prone, discouraging educators from conducting consistent analysis, which hinders improvements in test design and learning outcomes.

Optical Mark Reader (OMR) machines automate grading, they are costly and often out of reach for resource-limited schools, forcing educators to rely on manual methods that hinder data-driven teaching. Although OMR offers faster grading, it requires specialized equipment and maintenance, making it impractical for many. In contrast, machine learning methods like the K-Nearest Neighbors (KNN) algorithm provide a cost-effective alternative. KNN is simple, efficient, and can automate bubble sheet grading and item analysis, overcoming the limitations of OMR.

This study proposes using KNN with image processing to automate grading and generate detailed item analysis, offering an affordable, scalable solution for schools with limited resources, and improving the accuracy and efficiency of assessments.

Objectives

This research aims to develop **BubbleVision**, an automated system that leverages the K-Nearest Neighbors (KNN) algorithm for efficient grading and item analysis of bubble test sheets. The system seeks to offer an affordable, accurate, and accessible solution to help educators analyze student responses and improve test design. The specific objectives are:

(1) To design and develop a system that automatically identifies and classifies student responses on bubble test sheets using the K-Nearest Neighbors (KNN) algorithm. The system will process uploaded or photographed answer sheets, detect filled bubbles, and convert the data into a format that can be analyzed digitally.

(2) To create an automated item analysis feature that evaluates the difficulty level of test items and highlights questions identified as difficult, right difficult, and easy. The system will generate straightforward reports that help educators refine test items and improve assessment quality.

(3) To evaluate the performance of the BubbleVision system, focusing on its accuracy in answer detection, the reliability of item analysis, processing speed, and user satisfaction. The evaluation will assess the system's overall effectiveness in supporting educators in test analysis and item improvement.

Significance of the Study

This study is significant because it introduces a cost-effective, accurate, and efficient tool for conducting item analysis. By automating the process, teachers and schools can save time, minimize errors, and make better-informed decisions to improve teaching and learning.

This research will be helpful to:

Teachers – The system will help teachers quickly identify which test items are too easy, too difficult, or not effective, allowing them to improve the quality of their tests and teaching strategies.

Students – Students will benefit from fairer and better-quality assessments since teachers can revise or improve test questions based on item analysis results.

Educational Institutions – Schools can adopt the system as a cost-effective alternative to expensive optical mark reader (OMR) machines, making item analysis more accessible.

II. RELATED WORKS

Item analysis has become a vital procedure in enhancing the quality of school assessments. Instead of depending exclusively on raw scores, educators can assess the effectiveness of each test question by looking at the number of students who answered it correctly. This enables educators to pinpoint questions that might be too simple and thus not useful in assessing true learning. By identifying these elements—especially those labeled as easy, and difficult—educators can modify them to guarantee that evaluations more sincerely reflect student comprehension and teaching objectives. This approach improves the dependability of assessments and offers useful insights for refining test development.

Item Analysis in School Examinations

Studies on item analysis in education focus on improving the quality of assessments by examining difficulty, discrimination, and distractor efficiency. Researchers emphasize that analyzing test items not only enhances reliability but also provides feedback for instructional improvement and better measurement of student learning outcomes.

Siri and Freddano (2011) [1] analyzed school-level objective tests and showed that item analysis can effectively identify poorly constructed questions, helping teachers revise assessments to be more valid and reliable.

Sreekala and Arokia Marie (2015) [2] studied achievement tests for B.Ed. Physical Science students, finding that while most items were valid, some needed removal due to low discrimination indices.

Talebi et al. (2013) [3] applied item analysis in a physiotherapy course, highlighting that revising weak items and adjusting teaching strategies improved both item quality and student performance.

Rashwan et al. (2024) [4] conducted post-examination analysis of pediatric MCQs, combining statistical indices with Bloom's taxonomy, and demonstrated how item analysis can refine medical exam question banks.

Similarly, a study on distractor efficiency (2023) [5] found that plausible distractors strongly influence item difficulty and discrimination, suggesting that test designers must pay close attention to option quality.

In another classroom-focused study, Cizek and colleagues (2017) [6] discussed how teachers can use reliability and item analysis tools to evaluate self-made tests, reinforcing its role in everyday educational settings.

Recent approaches also integrate software and models. Siregar et al. (2024) [7] used the Iteman program to analyze Indonesian high school exams, revealing that several items failed validity and needed revision, proving the usefulness of digital tools in item diagnostics.

In higher education, Natarajan et al. (2016) [8] conducted item analysis of pharmacology MCQs and found that several items did not meet acceptable difficulty levels, recommending modifications for future use.

A Rasch model-based study (2022) [9] applied Item Response Theory (IRT) to Indonesian language exams and showed how advanced models can provide deeper insights beyond classical test theory.

Lastly, Washington University (2021) [10] provided a practical guide to interpreting item analysis reports, serving as a framework for educators to understand indices like difficulty and discrimination in practice.

Together, these works establish that item analysis is a powerful tool for enhancing test quality in both schools and higher education. By diagnosing weaknesses in test items, adjusting distractors, and applying statistical indices, educators can ensure fairer, more reliable, and valid assessments that align with learning outcomes.

K-Nearest Neighbors (KNN) in Image Processing

Studies on the K-Nearest Neighbors (KNN) algorithm emphasize its ongoing significance in image processing, pattern recognition, and educational technology. KNN, being a non-parametric and instance-based classifier, is appreciated for its ease of use, minimal computational expense, and excellent performance on organized visual datasets. Its ability to differentiate subtle variations in pixel intensities renders it especially apt for tasks like handwritten digit recognition, optical character recognition (OCR), and feature-driven image classification—uses that closely relate to automated bubble test evaluation.

A frequently referenced use of KNN is in the recognition of handwritten digits. The research Handwritten Digit Recognition Using K Nearest Neighbor Classifier (2019) showed that KNN effectively distinguishes grayscale digit images by analyzing neighboring pixel patterns, attaining solid baseline results on the MNIST dataset. This confirms KNN's ability to classify small, consistent image regions, like determining whether a bubble on a test sheet is shaded or unshaded

Nair and Kashyap (2020), in their study Classification of Medical Image Data Using K-Nearest Neighbor and Finding the Optimal K Value, examined the performance of KNN on low-contrast and noisy medical images. Their results indicated that choosing an ideal k value greatly improves classification accuracy, highlighting the necessity of appropriate parameter adjustment when utilizing KNN for image-related tasks. This understanding is directly applicable to bubble detection, where variations in lighting and scan quality can influence pixel values.

KNN has been utilized in optical character recognition as well. The study Using K-Nearest Neighbor in Optical Character Recognition (2012) explained how KNN effectively categorizes text characters by utilizing extracted image features. This shows the algorithm's effectiveness for tasks related to shape identification and pixel-density analysis—essential methods in identifying filled response bubbles.

The survey Handwritten Character Recognition Using k-Nearest Neighbors Classifier (2021) offers a wider view by examining multiple KNN-based recognition systems in different languages and scripts. The authors determined that KNN retains effective performance when combined

with suitable preprocessing and feature extraction techniques, underscoring its importance in lightweight image classification systems.

While not solely focused on images, Balaji and Chatterjee (2017) explore Performance Analysis of the K-Nearest Neighbors Algorithm for Classification, demonstrating how KNN's accuracy fluctuates based on distance metrics and feature representation. Their results highlight that KNN's effectiveness is greatly influenced by data preparation—crucial when differentiating shaded bubbles, which can differ in thickness, brightness, or shading uniformity.

Together, these studies highlight the effectiveness and adaptability of KNN in classification tasks. Although deep learning methods have gained more attention, KNN continues to be a robust, user-friendly option for settings with constrained computational capabilities. Its demonstrated capacity to categorize pixel-based patterns bolsters its use in automated bubble test evaluation, aiming to accurately distinguish shaded regions from unshaded ones. This research expands upon these foundational studies by utilizing KNN for bubble-sheet recognition and incorporating item analysis methods to develop a viable solution for educational evaluation.

The related literature shows that K-Nearest Neighbors (KNN) remains a viable and efficient algorithm for image classification tasks. Research on handwritten digit recognition, medical image categorization, and optical character recognition consistently demonstrates KNN's ability to effectively distinguish patterns based on pixel intensity and visual features. Studies on distance metrics and feature representation emphasize that KNN's accuracy can be enhanced with proper preprocessing, which is crucial when working with uploaded or photographed images. However, despite its success in general image processing, KNN has rarely been applied in automated assessment systems, particularly for bubble-sheet evaluations. Current item-analysis methods still rely heavily on manual verification or costly software, limiting accessibility for many schools. This research addresses this gap by using KNN to automatically identify and categorize answers on bubble sheets, categorizing items as easy, right difficult, or difficult. This approach offers a cost-effective solution, integrating machine-learning techniques with educational assessment needs, and helps schools with limited resources improve the quality and efficiency of their assessments.

III. METHODOLOGY

This research employed a quantitative experimental methodology, incorporating the systematic gathering and analysis of numerical information to evaluate manual verification against an automated KNN-based technique. The independent variable in this research is the method applied for evaluating student responses—namely the KNN-based system compared to conventional manual checking—whereas the dependent variables encompass the precision of response categorization, processing time needed, and rates of error. Actual bubble test sheets from the involved classes were initially examined manually to determine the accurate set of student responses, after which the KNN-based system was applied to the same sheets. The outcomes of both approaches were analyzed to assess if the automated system can deliver response classifications with accuracy that matches or surpasses manual verification while ensuring enhanced efficiency.

Instruments of the Study

The main tool utilized in this research was a KNN-driven automated bubble detection and tagging system created in Python, incorporating a complete Optical Mark Recognition (OMR) pipeline to analyze upload or photographed answer sheets. The system employed OpenCV to execute all necessary preprocessing tasks, such as converting to grayscale, reducing noise, applying thresholding, detecting contours, extracting bubbles, and aligning sheets, ensuring the precise isolation of each bubble area for analysis. The K-Nearest Neighbors classifier was implemented using Scikit-learn, comparing pixel-intensity features of extracted bubbles with labeled samples to determine if each bubble was shaded or unshaded. NumPy and Pandas were used to manage image characteristics, organize data, and map responses. A lightweight backend developed with an HTML and CSS user interface, enabled educators to upload answer sheets and obtain automated tagging results. By integrating the OMR pipeline, KNN classification, and an operational web interface, the system functioned as an all-encompassing and user-friendly tool for identifying student answers and tagging test items as easy, moderate, right difficult, or difficult.

Procedure

The creation of the KNN-based bubble test verification system involved a series of steps that

integrated image-processing methods and efficient machine learning approaches to ensure precise identification of shaded answers. Python was chosen as the main programming language because of its clarity and its vast ecosystem for tasks related to computer vision and pattern recognition. The preprocessing phase executed the typical OMR workflow, which included changing uploaded or photographed answer sheets to grayscale, minimizing noise, fixing distortions, aligning the layout of the sheet, and isolating bubble areas via thresholding and contour detection. These measures guaranteed that every bubble was uniformly retrieved and prepared for classification.

The KNN classifier was developed with Scikit-learn, offering a simple framework for executing instance-based learning ideal for categorizing bubble images. Rather than training deep learning layers, the system utilized labeled bubble samples that indicated shaded and unshaded states. Pixel-intensity characteristics were obtained from every bubble, and an ideal k value was chosen by means of iterative testing to enhance response-classification precision. After configuration, the KNN algorithm assessed each incoming bubble against the saved samples to identify if it was marked or unmarked. The categorized answers were subsequently compiled into a digital response form for every student.

Once all responses were identified, the system evaluated the number of students who answered each item correctly. Questions that most students answered correctly were automatically classified as "easy," while the least correctly answered questions were tagged as "difficult." A streamlined backend managed data uploads, image processing, classification execution, and report generation. Instructors interacted with the platform via a simple web interface created with HTML and CSS, allowing them to upload or photographed documents and receive automated results instantly. The system provided an efficient automated tool for recognizing student responses and tagging test items as easy, moderate right difficult, or difficult by combining the OMR pipeline with KNN classification and a user-friendly web interface. This method enhanced the speed, accuracy, and accessibility of test verification while remaining available to educational institutions with limited technological capabilities.

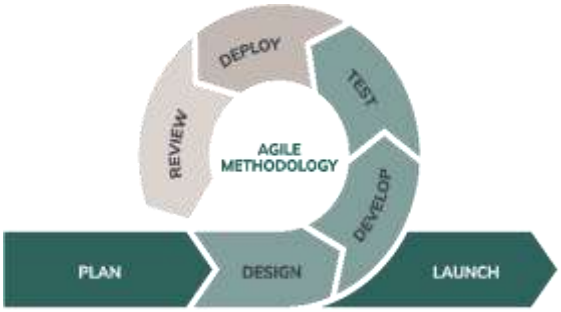

Fig. 1 Agile model

The study began with a clear planning stage focused on developing a KNN-based platform for automated bubble test evaluation and labeling of easy, moderate, right difficult and difficult items. At this phase, the research team pinpointed the crucial elements necessary as inputs, encompassing scanned bubble images, patterns of student responses, and collected accuracy rates. Conversations with teachers and school administrators clarified the project's scope, making sure that the system met actual classroom needs like decreasing manual verification time and automatically detecting items that were almost universally answered correctly by students.

During the design stage, these requirements were converted into a comprehensive system plan. The system encompassed the entire Optical Mark Recognition (OMR) preprocessing workflow with OpenCV, the KNN classification process for identifying filled bubbles, methods for data aggregation, and a streamlined web interface for educators. Wireframes and prototypes demonstrated the presentation of student answers and item tags, highlighting a user-friendly design that teachers could utilize without requiring technical skills.

Development was made via iterative Agile sprints, enabling the research team to construct and enhance system components incrementally. Early iterations concentrated on applying the OMR pipeline—transforming to grayscale, reducing noise, applying thresholding, detecting contours, and extracting bubbles—then proceeded to build the KNN model using labeled examples of both shaded and unshaded bubbles. Every sprint ended with internal assessments and feedback meetings with educators to verify that the developing system matched classroom conditions and user needs.

Testing was incorporated during development to confirm both the technical functionality of the KNN classifier and the platform's usability. The accuracy of bubble classification, the reliability of response

detection, processing speed, and error rates were analyzed and contrasted with the results from manual checks. The web interface was assessed to ensure that reports were straightforward, responsive, and simple to understand. Input from teachers influenced adjustments to bubble-detection limits, interface design, and tagging methods to enhance clarity and effectiveness.

The review phase aggregated all testing results to assess whether the platform achieved its established criteria for accuracy, reliability, and usability. Thorough evaluations verified that the system handled actual classroom data efficiently, ensured consistent response classifications across different image qualities, and accurately recognized test items that needed to be reviewed.

The system was deployed in a controlled school setting, where it underwent testing with real bubble sheets to assess scalability, compatibility with current workflows, and how easily it could be integrated into teachers' assessment practices. This stage confirmed that the KNN-based method operated efficiently in regular classroom environments.

Ultimately, the launch phase resulted in a completely functional and accessible bubble-checking platform prepared for wider application. Integrating the OMR image-processing pipeline with KNN classification and automated item tagging provided educators with a practical, efficient tool for simplifying test evaluations, improving assessment quality, and facilitating data-driven decisions in exam creation.

Methods of Evaluation for the K-Nearest Neighbors (KNN) System

To evaluate the performance of the KNN-based system as an automated tool for identifying shaded responses and marking items, multiple assessment methods were employed, emphasizing classification accuracy, system reliability, and practical applicability in an educational context. These procedures assessed the reliability of the KNN model in detecting shaded bubbles and evaluated its consistency in marking items that nearly all students answered correctly when applied to actual classroom data.

(1) Cross-Validation Strategy

The KNN classifier underwent assessment through repeated k-fold cross-validation on labeled bubble samples. This guaranteed that the classifier performed effectively across various groups of bubble images and did not rely excessively on shading patterns or sheet conditions.

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(2) Classification Metrics for Bubble Detection

Since KNN does not require training of neural networks, performance was assessed using bubble-classification metrics such as accuracy, precision, recall, and F1-score. These metrics assessed the accuracy with which the system identified shaded versus unshaded bubbles under different scan qualities, lighting conditions, and variations in handwriting. The accuracy of the final item tags was also checked through manual verification.

Accuracy

The overall correctness of the classifier:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

The correctness of predicted shaded bubbles:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

How well the system detects actual shaded bubbles:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score

The harmonic mean of precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where:

- **TP** = True Positive (correctly classified shaded bubble)
- **TN** = True Negative (correctly classified unshaded bubble)
- **FP** = False Positive (unshaded classified as shaded)
- **FN** = False Negative (shaded classified as unshaded)

These metrics allowed the study to determine how effectively the system classified student bubble responses before tagging items.

(3) Image Preprocessing and Feature Reliability Checks

Because KNN is highly dependent on the quality of input features, the results from the OpenCV preprocessing pipeline—like thresholder images, bubbles identified by contours, and segmentations of the bubbles—were examined manually to confirm that the system was accurately classifying clean and well-isolated bubble images. This interpretability phase confirmed that the classifier was receiving reliable patterns instead of noisy or corrupted inputs.

(4) Confusion Matrix Analysis

Confusion matrices were created to assess the classification effectiveness of the KNN system. These matrices emphasized the frequency with which the model accurately recognized shaded and unshaded bubbles, along with any patterns of misclassification. This analysis enabled the identification of whether specific bubble forms, shading levels, or image conditions led to consistent inaccuracies.

(5) Computational Efficiency and Scalability

The runtime performance of the system was assessed by measuring the processing speed for each test sheet, which included the duration spent on OpenCV preprocessing and KNN classification. Further evaluations included handling larger groups of answer sheets to assess if the system preserved consistent performance with heightened demands—an essential factor for institutions managing numerous classes or regular evaluations.

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(6) Addressing Variability in Shading Patterns

Due to the variation in shading intensity, thickness, and consistency of actual student responses, further assessment methods were implemented to evaluate the system's resilience to these discrepancies. Various threshold settings, bubble dimensions, and illumination conditions were implemented to guarantee that the KNN classifier stayed consistent and did not show preference for lightly or heavily shaded reactions. This assisted in verifying that the system could consistently label items as easy, moderate, right difficult, and very difficult regardless of the shading technique.

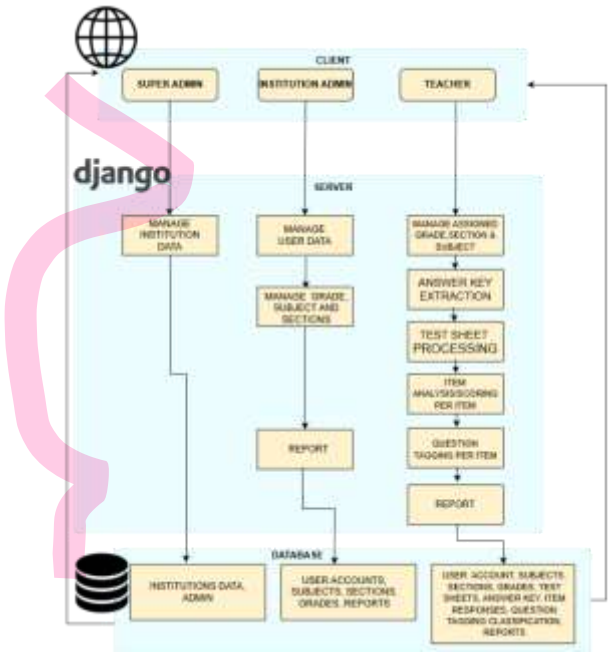

Fig 2. System Architecture

The Presentation Layer signifies the client aspect of the system, where the Super Admin, Institution Admin, and Teacher engage via a web-based interface. This interface is reachable through the internet and enables users to take actions according to their designated roles. The Super Admin oversees information related to the institution, whereas the Institution Admin manages user data and academic components like grades, sections, and subjects. Teachers engage with the system by overseeing their designated grades, sections, and subjects, submitting answer keys, and uploading or recording test sheets. This layer acts as the gateway for all user requests and showcases the reports generated by the system.

The Application Layer serves as the server side and is built using the Django framework. This layer handles all requests that are received from the client. It handles institutional data, user information, and educational details like grades, courses, and divisions. In the application layer, teachers can manage answer key retrieval, processing of test sheets, and analysis of test items. The system evaluates every test item, conducts scoring for each item, applies tagging to each question, and creates reports derived from the analyzed results. This layer manages all business logic, encompassing access control and data processing workflows, to guarantee precise and uniform system performance.

Finally, The Data Management Layer includes the database that retains all permanent system

information. This encompasses institution records, administrator information, user profiles, subjects, sections, grades, test papers, extracted answers, item responses, question tagging classifications, and produced reports. The application layer interacts with the database to safely access and save data, making sure that all system activities are accurately documented and preserved. This organized data storage facilitates dependable report creation, preserves historical archives, and allows the system to effectively handle numerous users and organizations.

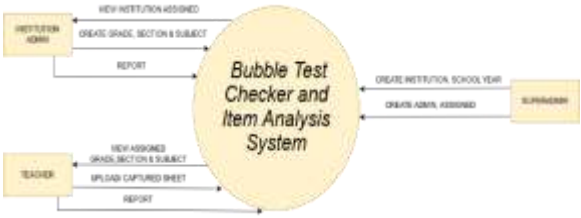

Fig. 3 Data Flow Diagram

The Data Flow Diagram (DFD) depicts the complete interaction between users and the Bubble Test Checker and Item Analysis System at the context (Level 0) level. This diagram depicts the complete system as one process that acquires inputs from various external entities and generates outputs in the shape of reports and processed data. The main objective of the system is to handle academic information, process bubble test forms, and produce item analysis outcomes to assist evaluation and decision-making in educational organizations.

The Super Admin acts as the top authority in the system and oversees system-wide configuration. This user supplies information like establishing institutions, outlining academic years, setting up administrator accounts, and designating institutional administrators. These functions guarantee that the system is correctly organized and prepared for various schools prior to the start of academic and evaluation activities.

The Institution Admin engages with the system by overseeing academic elements within their designated institution. This encompasses observing the institution they are assigned to, generating grades, sections, and subjects, and retrieving reports generated by the system. These interactions enable the Institution Admin to guarantee that academic data is correctly arranged and upheld in the system for efficient use by educators and administrators.

The system is mainly utilized by the Teacher for evaluation and analysis. Educators can check their allocated grades, classes, and topics, upload or take photos of bubble test sheets, and access created reports. These reports offer item analysis outcomes that assist teachers in assessing student performance and pinpointing areas requiring instructional enhancement.

In summary, the DFD illustrates that every data transfer among users is handled via the Bubble Test Checker and Item Analysis System, with no direct engagement between outside entities. The centralized data flow guarantees security, consistency, and accurate handling of test outcomes and reports, rendering the system an efficient instrument for automated evaluation and academic analysis.

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